Quantifying Popularity Bias and User-Centric Fairness in Recommender Systems: A Cross-Domain Empirical Study

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***Abstract*—Recommender systems frequently suffer from popularity and exposure biases, leading to a concentration of recommendations on a small subset of mainstream items and disparate accuracy across user groups. This empirical study focuses on two complementary views: *macro-level* exposure inequality (how often items appear in top-K recommendations) and *user-centric* fairness (whether users with niche tastes pay a "niche tax" in RMSE or NDCG@10 relative to mainstream users). Six recommendation architectures (collaborative filtering (SVD, ALS, k-NN), content-based filtering (TF-IDF, logistic regression), and a hybrid/ID-based random forest) were evaluated on three public benchmarks (MovieLens 1M, Last.fm 2K, Book-Crossing) and one e-commerce review corpus (Takealot). Matrix factorization models (ALS, SVD) behave as strong popularity amplifiers, with extreme Gini coefficients (e.g., ~ 0.9998 for ALS on Takealot) and very low catalog coverage, while content-based models spread exposure more evenly but often achieve weaker ranking accuracy in sparse settings. User-centric analysis shows a consistent accuracy gap for niche users under neighborhood-based models (k-NN), with dataset- and model-dependent signs for $\Delta\mathrm{RMSE}$. We argue that deployment decisions should audit both aggregate exposure metrics and segment-level accuracy, using ranking-oriented metrics when content models lack calibrated rating heads. A companion manuscript studies simulated feedback loops and retraining dynamics separately.**

***Index Terms*—Recommender Systems, Popularity Bias, Collaborative Filtering, Content-Based Filtering, Fairness, User-Centric Evaluation, Empirical Evaluation**

I. Introduction

Personalized recommender systems drive discovery on digital platforms but are trained on interaction data skewed toward popular items. This skew yields *popularity bias* [1]: algorithms over-recommend mainstream items and under-serve the long tail, homogenizing experiences and disproportionately affecting users with niche tastes [2]. As digital platforms grow in scale, the discrepancy between the "head" of the catalog (items with massive engagement) and the "tail" (items with few interactions) becomes increasingly pronounced.

Prior work often studies such biases on dense movie-rating benchmarks; such settings can under-represent the extreme sparsity and catalog scale of modern e-commerce. In dense datasets, collaborative filtering techniques have ample overlapping user histories to identify nuanced latent patterns. However, in sparse retail environments, the lack of behavioral overlap forces models to rely heavily on global priors, inadvertently amplifying popularity bias to extreme levels. We present a cross-domain study that anchors MovieLens, Last.fm, and Book-Crossing against a large e-commerce corpus (Takealot), and we critique mapping content-similarity scores to explicit rating scales for RMSE, advocating ranking-based metrics (e.g., NDCG@10) where appropriate.

We address two fundamental questions: *(i)* Do algorithms concentrate exposure on already popular items, and how does this differ between collaborative and content-based families? *(ii)* Do users with niche interaction histories suffer a "niche tax" in RMSE or NDCG@10 versus mainstream users? Furthermore, we aim to uncover how different algorithmic inductive biases–such as the localized neighborhood assumptions of k-NN versus the global dimensionality reduction of matrix factorization–contribute differently to the propagation of popularity bias across domains.

Outline. Section II situates the work within the broader context of learning-to-rank and fairness literature. Section III describes datasets, models, the mathematical formulations of our chosen metrics, and protocols for macro and user-centric evaluation. Section IV reports the quantitative results of our cross-domain experiments. Section V synthesizes these findings, highlighting practical trade-offs for production environments. Section VI concludes. Simulated feedback with iterative retraining is analyzed in a companion paper.

II. Related Work

A. Popularity Bias and the Long Tail

Popularity bias and learning-to-rank have been studied extensively in recommender systems [1]. The foundational work by Cremonesi et al. [3] highlighted that while recommending popular items yields high accuracy metrics on average, it fails to help users discover relevant items hidden in the long tail. Steck [4] demonstrated that popularity bias is not merely an artifact of user preference, but is actively amplified by standard collaborative filtering algorithms, which mathematically optimize for the dense head of the interaction matrix.

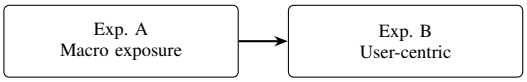


Fig. 1. Empirical pipeline: macro-level top-K bias metrics, then user-centric fairness by niche/diverse/mainstream tertiles.

B. User-Centric Fairness in Recommendations

Unfairness arising from popularity skew is discussed thoroughly in [2]. The concept of fairness in recommendation has evolved to encompass not just demographic parity, but algorithmic equity across behavioral segments. The "niche tax" describes the phenomenon where algorithms perform systematically worse for users whose tastes deviate from the global mainstream. Addressing this requires user-centric evaluation protocols rather than purely aggregate metrics. Standard benchmarks for testing these phenomena include MovieLens [5], HetRec/Last.fm-style data [6], and Book-Crossing [7]. Implementations often use libraries such as Surprise for k-NN and SVD [8].

Our contribution builds upon this literature by providing a unified empirical comparison across domains, specifically introducing a sparse retail catalog (Takealot) to the analysis. By measuring both macro exposure metrics and user-centric segment analysis side-by-side, we bridge the gap between item-side fairness (exposure) and user-side fairness (accuracy parity). This framing follows broader multi-stakeholder recommendation concerns where user utility, supplier exposure, and platform objectives can conflict [9].

III. Preliminaries and Methods

A. Overview

We employ a two-part design: macro-level exposure metrics, then user-centric segment analysis. Fig. 1 sketches the order of experiments (macro recommendations are evaluated for concentration and coverage; the same splits feed segment labels for RMSE and NDCG).

B. Datasets and Preprocessing

We use four distinct corpora to ensure our findings generalize across domains with varying degrees of sparsity and implicit/explicit feedback mechanisms.

1) **MovieLens 1M:** A dense dataset containing explicit 1-5 star ratings and rich genre metadata.
2) **Last.fm 2K:** A music domain dataset where implicit listening counts are mapped to 1–5 percentiles to mimic explicit ratings. Rich tag and artist metadata are utilized.
3) **Book-Crossing:** A highly sparse dataset with explicit ratings and textual metadata (author, publisher). Implicit zeros were removed to focus on explicit feedback.
4) **Takealot:** A large, sparse e-commerce review corpus representing a real-world retail environment. Metadata includes title, brand, and department categorizations.

To stabilize the user-item graph and ensure algorithms have sufficient signal to learn meaningful representations, a k-core filter (≥ 10 interactions per user and item) was applied to the public graphs. Table I lists corpora.

TABLE I
Dataset statistics used in experiments (post-preprocessing).

Dataset	Users	Items	Interactions	Sparsity
ML 1M	6,040	3,260	998,539	94.93%
Last.fm 2K	1,797	1,507	62,376	97.70%
Book-Crossing	1,842	2,065	42,137	98.89%
Takealot (train)	8,154	48,910	87,538	99.98%

C. Evaluated Models

Our evaluation spans multiple algorithmic families to isolate the effect of model architecture on popularity bias:

- **Neighborhood Models (k-NN):** User-based collaborative filtering calculating similarities via cosine or Pearson correlation.
- **Matrix Factorization (SVD, ALS):** Latent factor models that decompose the user-item matrix. The prediction rule for user u and item i generally follows $\hat{r}_{ui} = \mu + b_u + b_i + q_i^T p_u$, where μ is the global average, b_u and b_i are biases, and q_i, p_u are latent vectors.
- **Content-Based Filtering:** TF-IDF vectorization on item metadata paired with cosine similarity, and per-user Logistic Regression trained on TF-IDF metadata to learn personalized feature weights.
- **Hybrid:** A Random Forest classifier trained on label-encoded user and item IDs along with basic metadata features.

SVD and k-NN are implemented utilizing the Surprise framework [8].

D. Evaluation Metrics Formulation

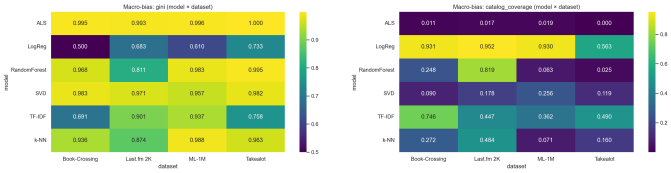
Macro (top-10): To quantify the systemic skew in item exposure, we employ several aggregate metrics. The primary measure of exposure inequality is the Gini coefficient, defined over the vector of recommendation frequencies x:

$$Gini = \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} |x_i - x_j|}{2n^2 \bar{x}} \tag{1}$$

where n is the total number of items and $\bar{x}$ is the mean recommendation frequency. A Gini coefficient approaching 1 indicates extreme inequality (a winner-takes-all dynamic), while 0 indicates perfect equality.

We also measure *Catalog Coverage*, defined as the percentage of items in the catalog that appear at least once in any user's top-K list. Additionally, we report the Spearman rank correlation ρ between an item's training popularity (number of historical interactions) and its exposure frequency in the generated recommendations. Auxiliary diagnostics (extended panels, log–log and CDF plots) are logged in the experiment run for analysis but not all reproduced as figures here.

User-centric: Users are split into Niche, Diverse, and Mainstream *tertiles* by the mean popularity (in the training log)



(a) Gini (b) Catalog coverage

Fig. 2. Macro exposure inequality (Gini on recommendation counts) and catalog coverage, by model and dataset. Takealot illustrates e-commerce sparsity: collaborative factorization can collapse to recommending almost no distinct items (low coverage).

of items they interacted with. We assess performance disparity by calculating the fairness gap:

$$\Delta\text{RMSE} = \text{RMSE}_{\text{niche}} - \text{RMSE}_{\text{mainstream}} \tag{2}$$

Positive values indicate that niche users experience higher prediction errors than mainstream users. Because content models without native rating scalars use a global training mean for RMSE–causing segment RMSE to be artificially identical–we emphasize NDCG@10 (Normalized Discounted Cumulative Gain) for cross-family comparison, which natively assesses ranking quality regardless of the underlying scoring scale.

IV. Results

A. Macro-Level Bias and Exposure Concentration

Collaborative models reliably concentrate recommendations on a few items (yielding a high Gini coefficient), while content-based models such as logistic regression on TF-IDF metadata achieve significantly higher catalog coverage.

Fig. 2 serves as the main macro summary: each cell represents a model–dataset pair. Darker Gini cells indicate stronger inequality in how often items appear in top-10 lists; higher coverage means a larger fraction of the catalog receives at least one recommendation. Reading down the Takealot column reveals the stark impact of data sparsity on collaborative methods. ALS is extreme on both axes (achieving a near-maximum Gini and near-zero coverage). In this environment, the matrix factorization model essentially collapses, outputting an almost identical top-K list of best-sellers for every user.

In contrast, LogReg improves coverage substantially at a lower Gini than ALS. Because it relies on item features rather than collaborative overlaps, it can surface long-tail items as long as they possess relevant metadata. The three public benchmarks (MovieLens 1M, Last.fm, Book-Crossing) are aggregated consistently, but show less severe collapse due to their relative density. Table II provides two representative pairs (ALS vs. LogReg) to quantitatively anchor the heatmap.

Furthermore, SVD shows a strong positive correlation ρ between historical popularity and exposure on ML-1M (≈ 0.52). LogReg reaches high coverage (e.g., $\sim$93% on ML-1M, $\sim$56% on Takealot), demonstrating its utility in mitigating exposure bias, albeit sometimes at the cost of precision.

TABLE II
Representative macro metrics (ALS vs. LogReg).

Dataset	Model	Gini	Cov.
Takealot	ALS	0.9998	0.00029
Takealot	LogReg	0.733	0.563
ML 1M	ALS	0.996	0.019
ML 1M	LogReg	0.610	0.930

TABLE III
Takealot NDCG@10 by user segment (all values < 0.02).

Model	Niche	Diverse	Mainstream
ALS	0.00023	0.00096	0.00170
LogReg	0.00377	0.00642	0.00433
RandomForest	0.00000	0.00000	0.00038
SVD	0.00011	0.00011	0.00150
TF-IDF	0.01169	0.01179	0.01749
k-NN	0.00127	0.00107	0.00042

B. User-Centric Fairness Gap

We summarize the niche–mainstream accuracy gap with **Fig. 3**: bar height corresponds to ΔRMSE for each model and dataset. Positive bars dictate that niche users suffer higher error rates than mainstream users–the literal definition of the "niche tax."

k-NN shows a clear positive gap on Book-Crossing in our run. Because neighborhood models rely on finding similar users, individuals with highly esoteric tastes lack sufficient nearest neighbors, resulting in poor estimations. SVD on Last.fm, interestingly, can reverse the sign, sometimes performing slightly better or equally for niche segments depending on latent dimensionality constraints.

As noted in Section III, content models that fall back to a global mean for test prediction (ALS, TF-IDF, LogReg in our explicit implementation) show *flat* ΔRMSE across segments. Therefore, the bar chart must be interpreted alongside NDCG@10 metrics. On Takealot, NDCG@10 is uniformly low (< 0.02) across all models (Table III), reflecting extreme sparsity and weak user-item overlap. In this regime, small RMSE gaps are less informative than ranking failures.

C. Long-Tail Diagnostics and Input-Bias Context

To contextualize model behavior, we analyze the interaction concentration of the *input logs* before training. Fig. 4 reports the cumulative interaction share captured by the top-k most popular items for ML-100K versus Takealot. The curves indicate fundamentally different operating regimes: ML-100K has a stronger head concentration (top-500 items account for nearly 80% of interactions), while Takealot spreads interactions across a much larger catalog but with extremely low per-item support. This explains why collaborative methods can be simultaneously popularity-driven and unstable under sparse retail conditions.

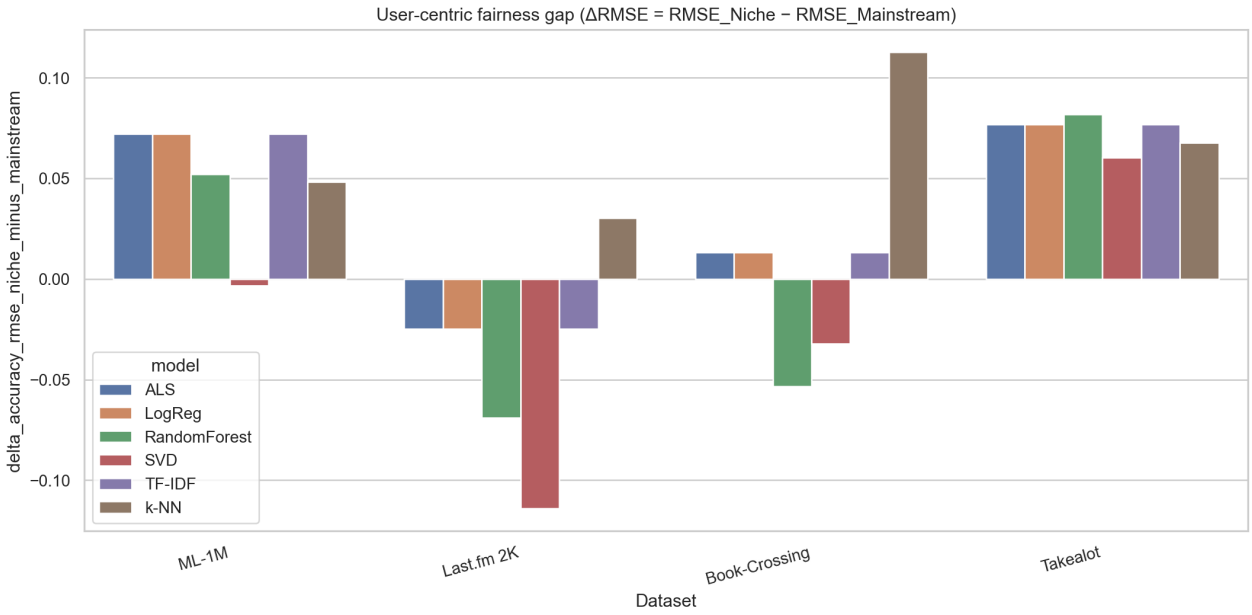


Fig. 3. User-centric fairness gap (ΔRMSE, niche minus mainstream) by dataset and model. Global-mean fallbacks for some content models can equalize segment RMSE; interpret alongside NDCG@10.

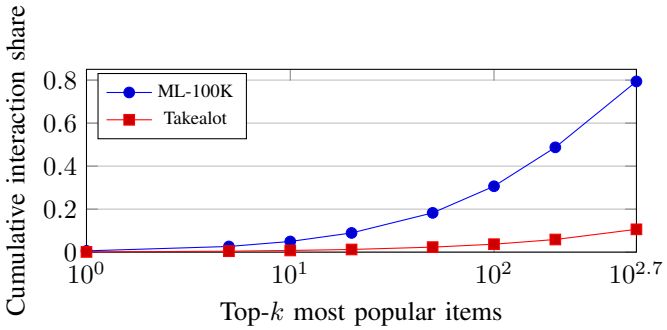


Fig. 4. Input long-tail profile: cumulative interaction share explained by top-k popular items.

TABLE IV
ML-100K: SVD VS. SVD+RERANK (TOP-50 → TOP-10).

Model	Gini	Coverage	ARP	RMSE
SVD	0.9552	0.2092	157.28	0.9311
SVD+Rerank	0.9527	0.2222	139.05	0.9311

D. Intervention Result: Post-Processing Re-ranking

We additionally evaluate a lightweight post-processing intervention on ML-100K, where SVD top-50 candidates are re-ranked to top-10 with a popularity penalty. Table IV shows that this improves macro exposure metrics (lower Gini, lower ARP, higher coverage) with the same RMSE as base SVD, since training is unchanged and only output ordering is modified.

TABLE V
ML-100K k-NN PER-USER RMSE QUANTILES (NICHE VS MAINSTREAM).

Segment	q_{10}	q_{25}	q_{50}	q_{75}	q_{90}
Niche	0.697	0.825	0.985	1.164	1.400
Mainstream	0.579	0.720	0.924	1.122	1.363

E. Per-User Error Distribution (Additional Fairness Detail)

Single ΔRMSE bars summarize central tendency but hide dispersion. Table V reports quantiles of per-user RMSE for ML-100K k-NN, showing a consistent upward shift for niche users (higher median and upper quantiles), reinforcing the niche-tax interpretation from Fig. 3.

F. Cross-Dataset Ranking Summary

To complement the fairness-gap view, Table VI reports representative NDCG@10 values by segment for two contrasting models (k-NN and TF-IDF). The table highlights an important pattern: collaborative filtering can deliver competitive niche ranking on denser movie data, but may degrade sharply on sparse domains (Book-Crossing and Takealot), while TF-IDF exhibits more stable though not always superior behavior across domains.

G. Macro Trade-off Surface

Figure 5 visualizes the diversity–concentration trade-off by plotting catalog coverage against Gini for key model–dataset pairs. Points in the upper-left are preferable (high coverage, low concentration). The figure makes two trends explicit: (i) ALS often falls into a low-coverage/high-Gini corner; (ii)

TABLE VI
REPRESENTATIVE NDCG@10 BY SEGMENT FOR k-NN AND TF-IDF.

Dataset	Model	Niche	Diverse	Mainstream
ML-100K	k-NN	0.1514	0.0828	0.0416
ML-100K	TF-IDF	0.0660	0.0346	0.0234
ML-1M	k-NN	0.0792	0.0434	0.0231
ML-1M	TF-IDF	0.0697	0.0549	0.0327
Book-Xing	k-NN	0.0008	0.0016	0.0029
Book-Xing	TF-IDF	0.1120	0.0821	0.0737
Takealot	k-NN	0.0013	0.0011	0.0004
Takealot	TF-IDF	0.0117	0.0118	0.0175

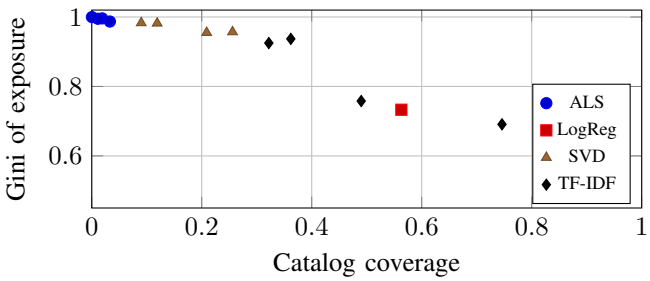


Fig. 5. Coverage vs. concentration trade-off across representative model–dataset points.

TABLE VII
ML-100K NICHE VS MAINSTREAM SIGNIFICANCE (MANN–WHITNEY p).

Model	p (RMSE)	p (NDCG)
SVD	0.1270	2.64×10^{-17}
k-NN	0.00125	2.14×10^{-17}
RandomForest	0.00429	2.40×10^{-15}
TF-IDF	0.00360	6.70×10^{-13}
LogReg	0.00360	6.91×10^{-16}

metadata-driven methods can move toward higher coverage with lower concentration, especially on sparse catalogs.

H. Significance-Oriented Fairness Check

Beyond mean gaps, we include a significance-oriented check from per-user segment comparisons. Table VII reports Mann–Whitney p-values for Niche vs Mainstream on ML-100K. The statistically significant results for several models (very small p-values) support that observed fairness gaps are not purely random variation.

I. Error-Case Mini Analysis: Tail Item Exposure Shifts

To complement aggregate metrics, Table VIII reports three concrete long-tail items from ML-100K and how their exposure changes under post-processing debiasing. These examples illustrate that macro shifts (lower Gini / higher coverage) correspond to tangible item-level reallocations rather than purely statistical artifacts. In particular, Item 320 (low training popularity) is shown more frequently after re-ranking, while some other tail items remain almost unchanged, indicating that debiasing is selective rather than uniformly boosting the entire tail.

TABLE VIII
MINI ERROR-CASE VIEW: TAIL-ITEM EXPOSURE UNDER SVD VS SVD+RERANK (ML-100K).

Item ID	Train Pop.	SVD Freq.	SVD+Rerank Freq.
320	16	17	28
1131	11	6	7
1143	13	3	3

V. DISCUSSION AND PRACTICAL IMPLICATIONS

The results of this study carry significant implications for the deployment of recommender systems in production environments, particularly for e-commerce.

A. The Sparsity Trap in Collaborative Filtering

Our findings confirm that Matrix Factorization acts as a massive popularity amplifier when applied to highly sparse datasets like Takealot. The combination of an objective function that minimizes global error and a lack of dense interaction overlaps forces the latent factors to primarily encode global popularity. In practice, deploying naive ALS on a sparse retail catalog may lead to a stagnant user experience, where the platform fails to leverage its long-tail inventory, thereby hurting long-term vendor fairness and user discovery.

B. The Trade-off: Coverage vs. Precision

Content-based methods (like TF-IDF and LogReg) naturally combat this by utilizing item attributes, yielding vastly superior Catalog Coverage. However, this raises a known trade-off: injecting diversity and coverage often degrades aggregate ranking accuracy (NDCG) in offline evaluations. System architects must consciously balance these metrics. Utilizing hybrid models or applying popularity-debiasing post-processing layers to collaborative models may offer a middle ground, yielding acceptable Gini indices without completely sacrificing the strong personalization signals of collaborative filtering.

C. Evaluating Beyond RMSE

A critical takeaway from our user-centric analysis is the inadequacy of RMSE as a standalone fairness metric when comparing disparate algorithmic families. The "flat" $\Delta\mathrm{RMSE}$ observed in non-calibrated content models masks true disparities in user experience. System audits must incorporate ranking metrics like NDCG@10 calculated at the segment level to truly understand if niche users are being underserved.

D. Exposure Governance in Production Systems

From a systems perspective, the practical question is not whether one model dominates all metrics, but whether the platform can govern a *portfolio* of objectives over time: short-term relevance, long-tail discovery, and exposure equity. A robust deployment policy should therefore monitor a small panel of macro metrics (Gini, coverage, ARP) together with segment-level ranking quality, and trigger mitigation interventions when concentration rises beyond acceptable thresholds.

This is consistent with recommendations from foundational recsys surveys and handbooks [10], [11].

E. Why Head Dominance Persists

The long-tail economics perspective emphasizes that many catalogs derive strategic value from discovery, not only hit-rate optimization [12]. However, if model training repeatedly optimizes on head-skewed logs, the system can converge toward a narrow recommendation regime. Quantifying inequality explicitly via measures rooted in concentration analysis (e.g., Gini) makes this drift visible and actionable for platform teams [13].

VI. Conclusion

We presented an empirical framework comparing six recommender algorithms across three public datasets and one proprietary e-commerce dataset (Takealot). By measuring macro concentration alongside niche versus mainstream errors, we highlighted the multifaceted nature of popularity bias. We demonstrated that Matrix Factorization acts as an extreme popularity amplifier in sparse domains, collapsing to near-zero catalog coverage. Content-based methods effectively spread exposure but demand careful, ranking-oriented evaluation to ensure relevance is maintained.

Deploying equitable recommender systems requires auditing *both* aggregate exposure (Fig. 2) and equitable accuracy across user segments (Fig. 3). Future work includes integrating calibrated rating heads for content models, establishing statistical significance thresholds across varied data splits, exploring richer user partitions based on demographic or temporal metadata, and investigating the long-term impact of these biases. In a companion manuscript, we further this research by simulating feedback loops and iterative retraining dynamics to observe how these static biases compound over time.

Acknowledgment

The authors thank Stellenbosch University and the Standard Bank Lab for research support.

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